1 Transpose of a Linear Map

Definition 1.1 (Transpose of a Linear Map)

Let V and W be vector spaces over F, and let $T \in \mathrm{Hom}(V, W)$. The *transpose of* T (or *dual map of* T) is the linear map $T^t \in \mathrm{Hom}(W^*, V^*)$ defined for $f \in W^*$ by:

$$T^*(f) = f \circ T.$$

That is, for each linear functional $f \in W^*$, $T^t(f)$ is the linear functional in V^* defined by $T^*(f)(v) = f(Tv)$ for each $v \in V$.

Exercise 1.2

Since $f \in W^* = \mathrm{Hom}(W, F)$ and $T \in \mathrm{Hom}(V, W)$, because the composition of linear maps we indeed have $f \circ T \in \hom(V, F) = V^*$ for each $f \in W^*$, which shows T^t indeed maps W^* to V^*. Show that $T^t : W^* \to V^*$ is linear.

Proof:

We have $T^t(\alpha f + g) = (\alpha f + g) \circ T$, and evaluating at a point gives:

$$(\alpha f + g)(T)(v) = (\alpha f + g)(Tv) = \alpha f(Tv) + g(Tv) = \alpha T^t(f)(v) + T^t(g)(v),$$

showing linearity.

Example 1.3

Consider the differentiation map $D : \mathcal{P}(\mathbf{R}) \to P(\mathbf{R})$ defined by $Dp = p'$. Fix $t \in \mathbf{R}$ and let $\mathrm{ev}_t \in \mathcal{P}(\mathbf{R})^*$ be the evaluation map which sends $p \to \mathrm{ev}_t(p) = p(t)$. Then

$$D^t(\mathrm{ev}_t)(p) = \mathrm{ev}_t(Dp) = p'(t) = p'(t).$$

Thus, $D^t(\mathrm{ev}_t) \in \mathcal{P}(\mathbf{R})^*$ is the linear function taking $p \to p'(t)$.

Exercise 1.4

Let D again be the differentiation operator on $\mathcal{P}(\mathbf{R})$. Let $\varphi \in \mathcal{P}(\mathbf{R})^*$ be the definite integral $\varphi(p) = \int_0^1 p(t)\,\mathrm{d}t$. What is $D^t(\varphi)$?

Solution

We have:

$$D^t(\varphi)(p) = \varphi(Dp) = \int_0^1 p'(x)\,\mathrm{d}t = p(1) - p(0).$$

Exercise 1.5

Let $\mathrm{id}_V \in \mathrm{Hom}(V, V)$ be the identity map on V. What is $\mathrm{id}_V^* t \in \mathrm{Hom}(V^*, V^*)$? Similarly, if 0_V is the zero map on V, what is 0_V^t?



Solution

We have $\mathrm{id}_V^t(f)(v) = f(\mathrm{id}_V(v)) = f(v)$, so $f \to f$, making $\mathrm{id}_V^t = \mathrm{id}_{V^*}$. We also have $0_V^t(f)(v) = f(0_V v) = f(0) = 0$ (for all linear maps), so 0^t is the map sending $f \to 0_{V^*}$.



Proposition 1.6 (Properties of the Transpose)

For all linear maps T_1, T_2, T and $\alpha \in F$,

(a) $(T_1 + T_2)^t = \alpha T_1^t + T_2^t$ (that is, $T \to T^T$ is linear)

(b) $(T_1 T_2)^t = T_2^t T_1^t$.



Solution

The former is obvious, since:

$$(\alpha T_1 + T_2)^t(f) = f((\alpha T_1 + T_2)(v)) = \alpha f(T_1 v) + f(T_2 v) = \alpha T_1^t(f)(v) + T_2^t(f)(v).$$

We have:

$$(T_2 T_1)^t(f) = f(T_2 T_1 v) = T_2^t f(T_1 v) = T_1^t T_2^t f(v),$$

so $(T_2 T_1)^t = T_1^t T_2^t$

2 Matrix of the Transpose of a Linear Map

Let V, W be finite-dimensional vector spaces. Choose ordered bases $B_V = (v_1, ..., v_n)$ and $B_W = (w_1, ..., w_m)$ of V and W. let $B_V^* = (f_1, ..., f_n)$ and $B_W^* = (g_1, ..., g_m)$ be ordered bases of V^* and W^*, respectively.



Proposition 2.1

The matrix of T^t is the transpose of the matrix T. That is,

$$[T^t]_{B_W^* + B_V^*}] = ([T]_{B_V, B_W})^t.$$



Exercise 2.2

Let V and W be finite-dimensional vector spaces, $T \in \mathrm{Hom}V, W)$ and $g \in W^*$. Choose ordered bases $B_V = (v_1, ..., v_n)$, and $B_W = (w_1, ..., w_m)$ for V and W respectively, and let $B_V^* = (f_1, ..., f_n)$ and $B_W^* = (g_1, ..., g_m)$ be the corresponding dual bases of V^* and W^*. Prove that $[T^t g]_{B_V^*, S} = [g]_{B_W^*, s}[T]_{V,W}$, where S denotes the standard ordered basis of F.



Remark 2.3

One can write the result of Exercise 2.2 in terms of columns vectors rather than row vectors by taking the transpose of both sides of $[T^t g] = [g][T]$, which gives $[T^t g]^t$, which gives $[T^t g]^t = [T]^t [g]^t = [T^t][g]^t$.